

Khwab Kalra

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EDUCATION

- **New York University Tandon School of Engineering** Brooklyn, NY
– M.S. in Computer Science (GPA: 4.0/4.0 Expected) Aug 2026 – May 2028
– **Coursework:** Distributed Systems, Machine Learning, Deep Learning, Data Structures, Computer Vision
- **National Institute of Technology Hamirpur** Himachal Pradesh, India
– B.Tech Electrical Engineering (GPA: 7.81/10) Aug 2019 – May 2023

TECHNICAL SKILLS

- **Languages:** Python (Expert), Java, C++, SQL, R, JavaScript, C
- **AI/ML & LLMs:** TensorFlow, PyTorch, Scikit-learn, LangChain, LangGraph, RAG, Prompt Engineering, Semantic Caching, Multi-Agent Systems
- **Cloud & MLOps:** AWS (Bedrock, SageMaker, ECS, Lambda, DynamoDB), Azure, Docker, Kubernetes, Terraform, FastAPI, CI/CD
- **Databases:** PostgreSQL, MySQL, MongoDB, Snowflake, Pinecone, DynamoDB, PgVector
- **Design:** Adobe Creative Suite, Canva, Figma (familiar), Graphic Design

EXPERIENCE

Ericsson

Noida, India

AI & Intelligent Automation Engineer

Nov 2023 – Present

- Architected enterprise AI/ML platform on AWS serving 50+ production services with 99.9% availability; automated CI/CD reducing deployment cycles from 3–4 weeks to **20 minutes**.
- Built financial agent harness using LangChain and AWS Bedrock; reduced manual reconciliation by **35%**.
- Engineered LangGraph-based multi-agent systems; reduced SME workload by **30%** (1,600 hours/year).
- Designed semantic caching layer; reduced LLM token costs by **25%**; implemented RBAC via AWS IAM.
- Built persistent conversation memory using DynamoDB; improved response accuracy by **20%**.

Oracle Financial Services

Remote

Software Engineering Intern

Jun 2022 – Jul 2022

- Developed and deployed 7 enterprise banking applications; optimized PL/SQL backend improving system resilience and data integrity.

NIT Warangal

Remote

Research Intern (Computer Vision)

May 2021 – Jul 2021

- Built deep learning object detection using TensorFlow and OpenCV; achieved **84.2% IoU** and **0.81 F1-score**.

PROJECTS

- **Enterprise RAG Agent with AWS Bedrock** — Built production RAG system with semantic caching; reduced token costs by **25%**. Python, AWS
- **High-Frequency Multi-Object Tracking** — Real-time tracking achieving **82.68%** accuracy; Kalman filtering. Python, C++, YOLOv4
- **I: APP — IoT Assistive System** — AI-powered IoT with hardware integration; improved efficiency by **25%**. Python, YOLOv7, Arduino
- **Marketing and Sales Analysis** — Regression analysis achieving **R2 score of 0.994**. Python, Pandas, Scikit-learn
- **HR Analytics Dashboard** — Interactive dashboard for workforce planning; processed large employee datasets. PowerBI, MySQL

PUBLICATION

- **Improved Deep Learning-based Contactless Biometric Recognition using Bracelet Lines**
– Proposed YOLO-based pipeline for biometric identification with statistical validation.

LEADERSHIP

- **Training and Placement Representative** — Liaison between 20+ corporate recruiters and 300+ students. 2021–22
- **Media and Marketing Head** — Led campaigns for 500+ participant festivals; managed budgets and vendors. 2022–23
- **Club Coordinator** — Led a 40-student team; mentored junior members. 2021–22
- **Core Coordinator** — Led a 30-member team for a virtual fest. 2022

AWARDS

- **\$400 Cash Award** — Ericsson AI/ML Innovation Challenge.
- **Top Performer** — Only candidate selected for Oracle Internship from class of 120.
- **GRE Quantitative:** Perfect Score (170/170).